

Project 12 — Computational Biology Capstone

An integrated computational biology workflow from biological question to interpretation

Research Question

How can sequence, structural, computational, and biological evidence be integrated into a single computational biology investigation?

Project Overview

This final capstone brings together the portfolio's sequence, structural, transcriptomic, programming, functional-analysis, and reproducibility concepts into a research-oriented framework.

Research Philosophy

```
Biological question
↓
Data collection
↓
Computational analysis
↓
Quality checking
↓
Biological interpretation
↓
Limitations
↓
Next research question
```

Integrated Workflow

```
Biological question
↓
Sequence / structural / expression data
↓
Quality control
↓
Sequence analysis
↓
Structural analysis
↓
Comparative analysis
↓
Functional interpretation
↓
Biological hypothesis
↓
Validation / future analysis
```

Sequence and Structural Analysis

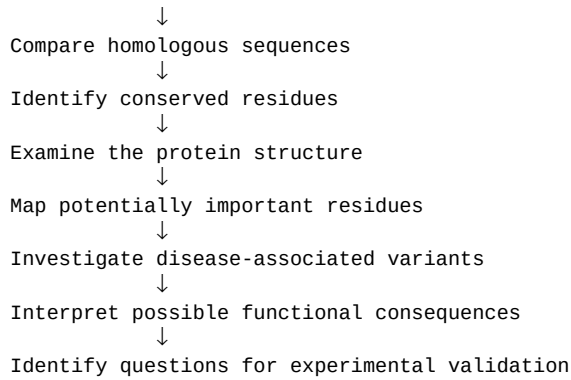
Sequence analysis can include FASTA processing, BLAST similarity, multiple sequence alignment, conservation, and phylogenetic relationships. Structural analysis adds three-dimensional context such as fold, RMSD, conserved structural regions, and mutation location.

Functional Interpretation

Depending on the dataset, functional interpretation can involve Gene Ontology, KEGG, interaction networks, conserved regions, disease-associated variants, or expression patterns. The method should follow the biological question.

Example Integrated Logic

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Identify a biologically relevant protein
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Scientific Interpretation

A strong analysis distinguishes observation, interpretation, and hypothesis. Observation is what the computation directly shows; interpretation is what it may suggest; a hypothesis is what should be tested through further analysis or experiment.

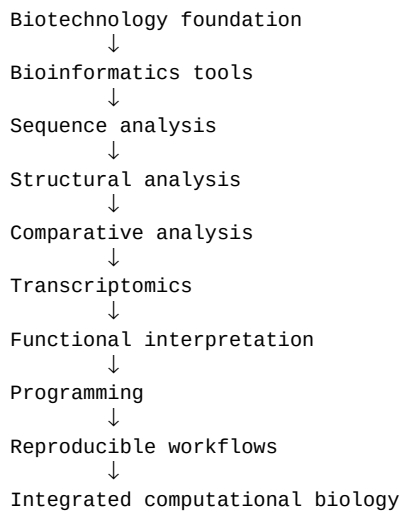
Reproducibility

A research-quality computational project should document dataset accessions, input files, software, parameters, computational steps, results, figures, interpretation, and limitations.

Limitations

Computational predictions do not automatically establish causation. Database annotations may contain uncertainty. Small datasets can produce unstable conclusions. Structural models represent limited conformational states, and experimental validation remains important for many hypotheses.

Portfolio Progression



Future Research Direction

The next stage is to move from educational portfolio projects toward independently reproducible research, potentially involving RNA-seq, variant annotation, structure prediction, molecular dynamics, network biology, machine learning, multi-omics, computational epidemiology, or health informatics.

Final Conclusion

The capstone represents a transition from learning individual bioinformatics tools toward thinking like a computational biology researcher: start with a biological question, choose an appropriate method, interpret results biologically, and communicate limitations clearly.